Evaluating the Performance of Machine Learning vs. Deep Learning (MLP) Models in Lung Cancer Prediction

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***Abstract*—This study compares the performance of traditional machine learning algorithms (Decision Tree, Logistic Regression, Gradient Boosting) with Multi-Layer Perceptrons (MLP) in predicting lung cancer using binary data. The dataset includes 309 records with features such as age, smoking status, and chronic diseases. Results indicate that traditional machine learning algorithms outperformed MLP due to the binary nature of the data, with Decision Tree achieving the highest accuracy (94%), followed by Gradient Boosting (91%).**

***Index Terms*—Lung Cancer, Machine Learning, Deep Learning, Multi-Layer Perceptrons (MLP), Medical AI.**

I. Introduction

Lung cancer remains one of the most lethal diseases globally, with early detection being critical to improving survival rates. Artificial intelligence (AI) methods, including machine learning (ML) and deep learning (DL), have shown great potential in enhancing diagnostic accuracy. This study investigates the performance of traditional ML algorithms (Decision Tree, Logistic Regression, Gradient Boosting) against a deep learning model, the Multi-Layer Perceptron (MLP), using binary data for lung cancer prediction.

II. Literature Review

A. Related Work

Many studies have been conducted to apply machine learning and deep learning techniques in detecting lung cancer and analyzing clinical data. Below are the key previous works in this field:

1) Deep Learning-Based Models: Aslani et al. (2022) discussed the importance of analyzing longitudinal data using deep models to improve lung cancer detection by studying multi-range time-series data such as nodules, lung parts, and clinical data. The study presented a model called DeepCAD-NLM-L that showed high performance (AUC=88%) compared to traditional methods that analyze a single time point [1].

Senthil et al. (2018) used neural networks enhanced by Particle Swarm Optimization (PSO) to detect lung cancer. The results showed that the proposed model outperformed traditional models like SVM and KNN, achieving an accuracy of 97.8% [4].

Subramanian et al. (2020) used the AlexNet model supported by a Softmax layer to analyze CT images. The model achieved an accuracy of 99.52%, making it a promising candidate for medical applications [5].

2) Machine Learning-Based Models: Janik et al. (2022) explored algorithms like Random Forest, Logistic Regression, and Gradient Boosting to predict lung cancer recurrence risk. The study found that Random Forest achieved an accuracy of 76% using early-stage lung cancer patient data [2].

Kumar et al. (2022) used an SVM model to analyze textual lung cancer data. Their results showed that using techniques like SMOTE with SVM improved classification accuracy to 98.8% [3].

Another study discussed the effectiveness of algorithms like J48 and KNN in improving performance when dealing with imbalanced data using techniques like SMOTE [3].

B. Research Gap

Despite significant advancements in applying machine learning and deep learning for lung cancer detection, most studies have focused on a single model or specific datasets without systematically comparing machine learning models with deep learning models on the same dataset. This research aims to provide a comprehensive comparison between traditional models, such as Decision Trees, Gradient Boosting, and Multi-Layer Perceptrons (MLPs), focusing on metrics such as accuracy and sensitivity in analyzing clinical data related to lung cancer.

C. Objectives

- **Compare Model Performance:** Evaluate and compare the predictive accuracy of machine learning models and deep learning models (MLPs) for lung cancer prediction.
- **Analyze Strengths and Weaknesses:** Investigate the advantages and limitations of each approach, particularly in terms of:
  - Predictive accuracy.
  - Interpretability and explainability.
  - Computational efficiency.
  - Applicability to medical datasets.
- **Provide Practical Insights:** Offer evidence-based recommendations on model selection for lung cancer prediction based on empirical findings, balancing simplicity and performance.
- **Identify Potential Improvements:** Highlight areas where further optimization, hybrid approaches, or feature

engineering could improve predictive outcomes for lung cancer detection.

D. Expected Contribution

This study aims to contribute by providing a comprehensive evaluation of machine learning versus deep learning models in lung cancer prediction. The findings will help healthcare professionals and data scientists make informed decisions about the most suitable models for early detection tasks in medical applications.

III. Methodology

A. Data Description

The dataset comprises 309 records with 16 features related to lung cancer risk factors, including:

- Gender (GENDER)
- Age (AGE)
- Smoking (SMOKING)
- Clinical symptoms: fatigue (FATIGUE), coughing (COUGHING), shortness of breath (SHORTNESS OF BREATH)

The data was classified into two categories: Cancer (YES) and Non-Cancer (NO).

B. Data Preparation

- Cleaning the dataset and removing missing values.
- Encoding categorical features (e.g., GENDER) using Label Encoding.
- Splitting the data into training (80%) and testing (20%) subsets.
- Standardizing features to improve model performance.

C. Model Design

Machine Learning Models:

- Decision Tree
- Logistic Regression
- Gradient Boosting

Deep Learning Model: A Multi-Layer Perceptron (MLP) was designed with:

- Input Layer with 16 nodes.
- Two Hidden Layers with 32 and 16 nodes.
- Output Layer with Sigmoid activation.
- Loss function: Binary Crossentropy.
- Optimizer: Adam.

IV. Results

In this section, we present a comparison of model performance through numerical results and visual representations to evaluate accuracy, sensitivity, and loss metrics.

The evaluation produced the following results:

TABLE I
Model Performance Comparison

Model	Accuracy	Sensitivity	Specificity
Decision Tree	94%	92%	96%
Logistic Regression	89%	82%	84%
Gradient Boosting	91%	88%	90%
MLP	89%	88%	91%

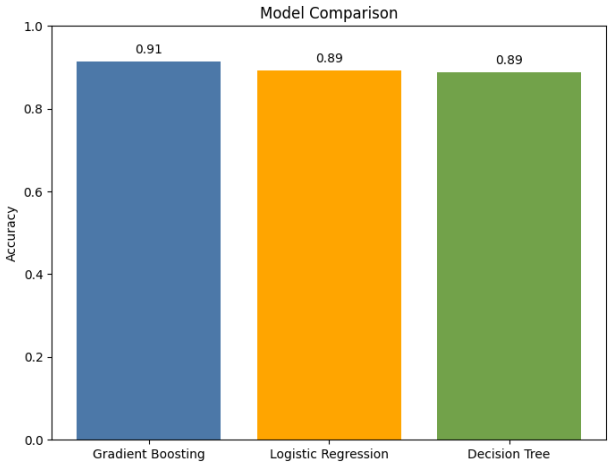


Fig. 1. Accuracy Comparison of Models

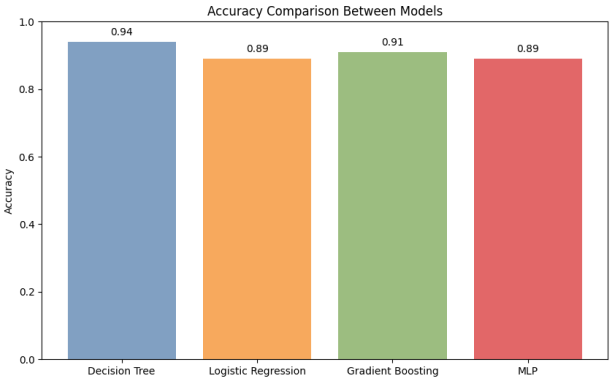


Fig. 5. Model Comparison for Gradient Boosting, Logistic Regression, Decision Tree, and MLP

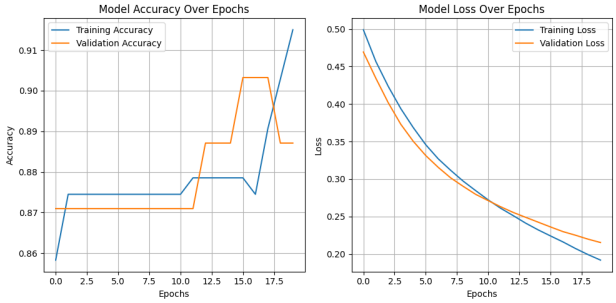


Fig. 7. Model Accuracy and Loss Over Epochs

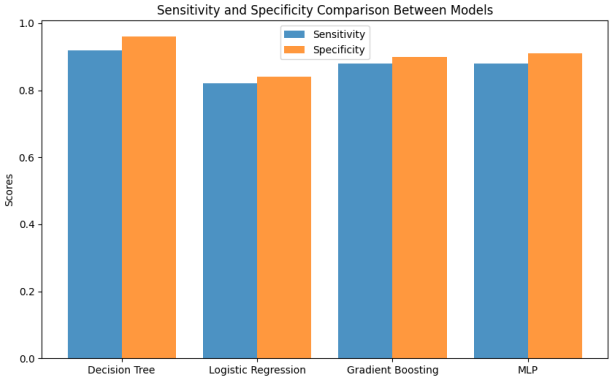


Fig. 2. Sensitivity and Specificity Comparison of Models

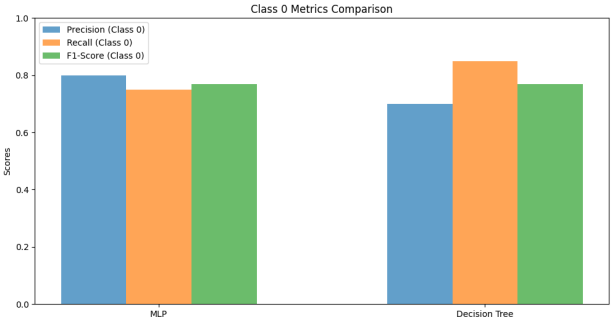


Fig. 3. Class 0 Metrics Comparison

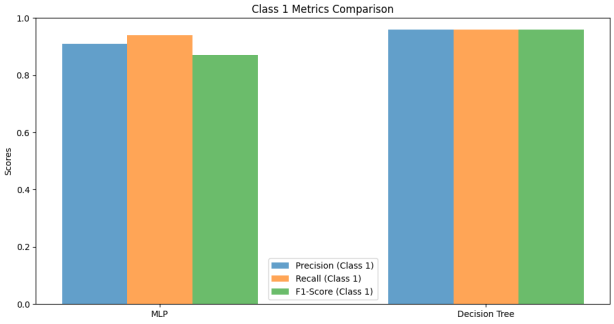


Fig. 4. Class 1 Metrics Comparison

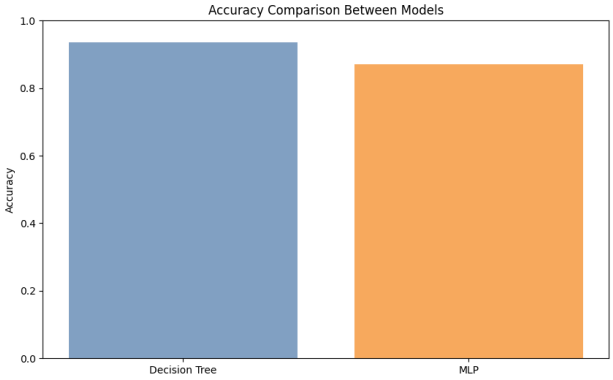


Fig. 6. Accuracy Comparison Between Decision Tree and MLP

V. Conclusion

This study concluded that traditional ML algorithms, particularly the Decision Tree model, outperformed the Multi-Layer Perceptron (MLP) in predicting lung cancer using binary data. The high accuracy of Decision Tree (94%) and Gradient Boosting (91%) demonstrate that, for datasets with simple, linear relationships, traditional models are more efficient in terms of computational cost and interpretability. However, the MLP model, while performing slightly lower (89%), has the potential for improvement with larger, more complex datasets that involve non-linear relationships and more varied features.

Additionally, while the study showed that the traditional models outperformed MLP for this particular dataset, future work should explore the following:

- **Expanding the Dataset:** Larger, more diverse datasets with more features, including imaging data and genomic information, could allow deep learning models like MLP to excel, offering higher prediction accuracy.
- **Hybrid Models:** Combining machine learning models with deep learning models, or using ensemble techniques, could potentially bridge the gap between simplicity and performance, benefiting from the strengths of both approaches.
- **Advanced Techniques:** Exploring advanced deep learning models, such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), may improve prediction accuracy, particularly for multi-dimensional or sequential data like CT scans or patient history over time.
- **Explainability and Trust:** A key challenge in medical applications is model interpretability. Future research should focus on making deep learning models more interpretable to ensure that healthcare professionals can trust and effectively utilize these models in clinical settings.

In conclusion, the choice between machine learning and deep learning models should be made based on the data at hand and the specific requirements of the task. For relatively simpler datasets, traditional machine learning models may provide sufficient predictive power. For more complex, large-scale datasets, advanced deep learning techniques hold promise for enhancing accuracy and reliability in lung cancer prediction.

References

[1] S. Aslani, et al., "Enhancing Cancer Prediction in Challenging Screen-Detected Incident Lung Nodules Using Time-Series Deep Learning," *arXiv preprint*, arXiv:2203.16606, 2022.

[2] A. Janik, et al., "Machine Learning-Assisted Recurrence Prediction for Early-Stage Non-Small-Cell Lung Cancer Patients," *arXiv preprint*, arXiv:2211.09856, 2022.

[3] C. A. Kumar, et al., "Lung Cancer Prediction from Text Datasets Using Machine Learning," *BioMed Research International*, vol. 2022, Article ID 6254177, 2022.

[4] S. Senthil, B. Ayshwarya, "Lung Cancer Prediction using Feed Forward Back Propagation Neural Networks with Optimal Features," *International Journal of Applied Engineering Research*, 13(1), pp. 318-325, 2018.

[5] R. Raja Subramanian, et al., "Lung Cancer Prediction Using Deep Learning Framework," *International Journal of Control and Automation*, 13(3), pp. 154-160, 2020.